GG-M-001130(GSA-FSS)
November 10, 1966

INTERIM FEDERAL SPECIFICATION

MARKER, TUBE TYPE, FINE TIP

This Interim Federal Specification was developed by the Standardization Division, Federal Supply Service, General Services Administration, Washington, D. C. 20406, based upon currently available technical information. It is recommended that Federal agencies use it in procurement and forward recommendations for changes to the preparing activity at the address shown above.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers a tube type, fine tip marker for use in freehand marking, writing, drawing, and identification marking on paper and porous surfaces.

1.2 Classification.

1.2.1 Colors of ink. The markers shall be furnished with inks of the following colors, as specified (see 6.1):

Black	Orange
Blue	Yellow
Red	Brown
Green	Purple

2. APPLICABLE DOCUMENTS

2.1 The following specifications and standards, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Specifications:

UU-P-121 - Paper, Bond and Writing, White and Colored.
PPP-B-566 - Boxes, Folding, Paperboard.
PPP-B-636 - Box, Fiberboard.
PPP-B-665 - Boxes; Paperboard, Metal Stayed (Including Stay Material).
PPP-B-676 - Boxes, Setup.

Federal Standard:

Fed. Std. No. 123 - Marking for Domestic Shipment (Civilian Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C., 20402.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D.C., Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, and Seattle, Wash.

FSC 7520

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Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Bid samples. Each bidder shall submit with his bid 12 markers of each color bid upon for the purpose of determining compliance with the characteristics listed in 4.2. The bid samples shall be in strict conformance with the markers the bidder proposes to furnish under contract and shall be capable of meeting all the requirements of this specification.

3.2 Hazardous substances. The marker ink shall contain no quantities of irritants, toxics, or flammable substances that may cause personal harm as a result of any reasonably foreseeable handling or use, including reasonably foreseeable ingestion by children.

3.3 Materials.

3.3.1 Plastic. Plastic used shall retain its design contours without warping, crazing, cracking, softening, or discoloring in service or storage and shall be nonflammable or shall have a burning rate not exceeding 3 inches per minute when tested as specified in 4.3.4.2.1.

3.3.2 Metal. Metal used shall be of a material appropriate for the intended purpose and when used externally shall be suitable for coating as required by 3.6.

3.4 Design and construction. The marker shall be a non-refillable type, designed to operate by capillary action. The marker shall be constructed to provide good protection against evaporation and leakage. The ink shall be water-based, shall be free from obnoxious odors, and shall be held within the marker by an absorbent material so that there is no free liquid within the marker. The marker shall be equipped with a tight-fitting plastic or metal protective cap and the over-all length of the marker with cap in place shall be not less than 5-1/2 inches. The cap or barrel shall be equipped with a plastic or metal pocket clip.

3.4.1 Marker barrel. The marker barrel shall be made of plastic or metal or a combination of these materials. The barrel diameter shall be not less than 0.310 inch, excluding tapered and recessed ends, and not more than 0.500 inch, and shall be constructed so that the ink will emerge only through the writing nib.

3.4.2 Writing nib. The writing nib may be of any material suitable for the intended purpose. The nib shall be permanently secured in position.

3.5 Performance characteristics.

3.5.1 Drying time. The writing shall not smear and the paper which is drawn across the writing shall not be smudged when tested as specified in 4.3.4.1.1.

3.5.2 Non-transferability. The markings shall not show a transference to the cover paper when tested as specified in 4.3.4.1.2.

3.5.3 Resistance to water. The letters shall retain their color identity and the words shall remain legible when tested as specified in 4.3.4.1.3.

3.5.4 Writing characteristics. There shall be no feathering, spreading, or splitting of the markings nor any penetration of ink to the reverse side of the paper when tested in accordance with 4.3.4.1.4.

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3.5.5 Strength of nib. The writing nib shall not break, crack, split, spread, or fray under the load and the mark shall have a line width in the 0.015 to 0.050 inch range when tested as specified in 4.3.4.1.5.

3.5.6 Pocket clip tension. The clip shall have sufficient tension to support the marker in the inverted position when tested as described in 4.3.4.1.6.

3.5.7 Startability. The marker shall produce a solid continuous mark not less than 0.015 inch wide within four strokes after laying with the cap off for three hours and within four strokes after laying with the cap off for 15 hours and recuperating for 30 minutes in accordance with 4.3.4.1.7.

3.5.8 Accelerated aging. The marker shall not leak or show visible distortion and the markings shall not show a change in color when tested as specified in 4.3.4.1.8.

3.5.9 Write-out distance. The marker shall write a minimum distance of 2500 feet when tested as specified in 4.3.4.1.9.

3.5.10 Light fastness. The markings shall be clearly visible and shall retain their color identity when tested in accordance with 4.3.4.1.10.

3.5.11 Body and cap tightness. The caps shall show no impairment and shall continue to fit securely on the marker barrels, and the average weight loss per marker shall not exceed 1.0 mg. when tested as specified in 4.3.4.1.11.

3.5.12 Impact resistance. There shall be no evidence of leakage, the cap or barrel of the marker shall show no impairment, and the writing nib shall show no signs of loosening when tested as described in 4.3.4.1.12.

3.6 Finish. The cap or body of the marker shall be colored, or shall have a colored marking, to indicate the color of ink content. All externally exposed metal shall be coated with a decorative, protective finish. The surface of the marker and all components shall have a smooth, uniform appearance and shall be free from cold bonds, delaminations, porosity, warpage, cracks, checks, chips, rough or sharp edges, burrs, blisters, or scratches.

3.7 Identification marking. The markers shall be legibly and permanently marked with the manufacturer's name or trademark.

3.8 Shelf life. The markers shall be able to meet all the requirements of this specification from the time of delivery to a minimum of 12 months in storage. Suppliers shall replace, without cost to the Government, all markers failing to meet this requirement.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Bid samples. Bid samples submitted in accordance with 3.1 will be examined for:

Balance.
Facility of use.
Comfortable grip.
Suitability of materials used.
Suitability of ink color.
Odor of ink.
Noise and drag of nib while writing.
General appearance and workmanship.

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In addition, bid samples will be subjected to the following tests to assure compliance with the corresponding requirements:

4.3.4.1.3 - Water resistance test.
4.3.4.1.8 - Accelerated aging.
4.3.4.1.9 - Machine write-out test.
4.3.4.1.10 - Light fastness.
4.3.4.1.12 - Impact test.

4.3 Sampling and inspection.

4.3.1 Lot. For purposes of sampling for inspection, a lot shall consist of all markers of one color offered for acceptance at one time.

4.3.2 Inspection of marker. Examine sample markers for the defects listed in table I. Sampling shall be in accordance with MIL-STD-105. The inspection level shall be S-3 with an acceptable quality level (AQL) of 4.0 defects per hundred units. The sample unit (unit of product) shall be one marker.

TABLE I. Inspection for marker defects.

Examine	Defects
Materials	Marker body and cap not plastic or metal or a combination of these materials; materials unsuitable for the purpose intended; ink gives off an obnoxious odor.
Design and construction	Marker not non-refillable type; other than capillary action feed; cap not tight-fitting; over-all length less than 5-1/2 inches; evidence of leakage.
	Pocket clip missing.
	Barrel diameter less than 0.310 inch (excludes tapered and recessed ends) or more than 0.500 inch.
	Writing nib not permanently secured in position.
Ink color	Not as specified (see 6.1).
Finish	No color on cap or body to indicate color of ink content; externally exposed metal not coated with a decorative, protective finish; surface (including components) not free from cold bonds, delaminations, porosity, warpage, cracks, checks, chips, rough or sharp edges, burrs, blisters, or scratches.
Identification marking	Marker not legibly and permanently marked with the manufacturer's name or trademark.

4.3.3 Inspection of preparation for delivery. An inspection shall be made to determine that the packaging, packing, and marking comply with the requirements of section 5. Defects shall be scored in accordance with table II. For examination of interior packaging the sample unit shall be one shipping container fully prepared for delivery, selected at random just prior to the closing

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operations. Sampling shall be in accordance with MIL-STD-105. Defects of closure listed shall be examined on shipping containers fully prepared for delivery. The lot size shall be the number of shipping containers in the end item inspection lot. The inspection level shall be S-2 with an AQL of 4.0 defects per hundred units.

TABLE II. Inspection for preparation for delivery defects.

Examine	Defects
Markings (exterior and interior)	Omitted; incorrect; illegible; improper size, location, sequence, or method of application.
Materials	Any component missing or damaged.
Workmanship	Inadequate application of components such as incomplete closure of container flaps, loose strapping, or distortion of container.
Contents (exterior and interior container)	Number per container is more or less than required. Net weight exceeds requirements.

4.3.4 Tests. Sample markers shall be subjected to the following tests. Sampling for the performance characteristics tests shall be in accordance with MIL-STD-105, inspection level S-1, with an AQL of 6.5 defects per hundred units. The sample unit shall be six markers: one marker for tests 4.3.4.1.1 through 4.3.4.1.6; one for test 4.3.4.1.7; one for test 4.3.4.1.8 and 4.3.4.1.9; and three for tests 4.3.4.1.11 and 4.3.4.1.12. The sample size shall be the number of sample units. The failure of a marker to pass one test shall constitute a defect. In selecting the samples, take not more than three markers from each package of 12.

4.3.4.1 Performance characteristic tests. Unless otherwise specified, the tests shall be conducted in an atmosphere of 23° ± 1.1° C. (73.5° ± 2° F.) and at a relative humidity of 50% ± 4%. Tests involving writing, except 4.3.4.1.9, shall be conducted with the markers held at an angle approximately 50° from the horizontal and with a writing pressure so that the markings are clearly visible. All test paper used shall conform to UU-P-121, type III, white (17 x 22 - 1000 sheets, 32 pound weight), with a maximum smoothness of 30 seconds on the felt side. The machine write-out test in 4.3.4.1.9 shall be conducted on an Anja Engineering Corporation writing machine, model W-10, modified for marker testing with Anja's adaptor kit; or equal. A carbon-arc lamp such as the Fadeometer, type FDA-R, or its equivalent, shall be used in the light-fastness test in 4.3.4.1.10.

4.3.4.1.1 Drying time. Place a sheet of paper on a smooth, flat surface and partially cover with a second sheet of paper. Place a 200 gram weight, approximately 1-1/4 inches in diameter, on top of the second sheet. Write a five-letter word on the exposed portion of the bottom sheet, allow to dry 3 seconds, and immediately draw the top sheet and the weight across the writing at a rate of approximately 4 inches per second. Examine the writing and top sheet of paper for compliance with 3.5.1.

4.3.4.1.2 Non-transferability. Write 4 five-letter words on a paper panel (use marker tested in 4.3.4.1.1) and allow to dry for 5 minutes. Then cover the markings with 3 layers of paper and place a rectangular, 4 x 6 inch piece of 1/4 inch plate glass flat on top of the test assembly. Place the entire assembly on a flat surface in a convection-type oven at a constant temperature of 80° ± 1° C. (176° ± 1.8° F.) for one hour. Remove the assembly from the oven and examine the cover paper for compliance with 3.5.2.

4.3.4.1.3 Water resistance test. Write 4 five-letter words (use marker previously tested) on a sheet of paper, allow to dry for 5 seconds, and immerse the paper in distilled water for 30 seconds. Then remove from water and check for compliance with 3.5.3.

4.3.4.1.4 Writing test. Using the marker from the preceding test write 4 five-letter words on a sheet of paper. After 3 minutes examine the markings and the reverse side of the paper and check for compliance with 3.5.4.

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4.3.4.1.5 Strength test. Place a sheet of paper on a platform scale. Grip the uncapped marker (use marker tested in preceding tests) between 1-1/4 and 1-1/2 inches from the writing end. Hold the writing nib to the paper in a normal writing position and at an angle of 50° from the horizontal without the hand touching the paper. Apply a force of 700 grams gradually and uniformly, observing the effects on the nib. Immediately release after 5 seconds of full pressure. Draw one 6-inch linear mark with the tested marker and examine for compliance with 3.5.5.

4.3.4.1.6 Pocket clip test. Using the marker tested previously, slide the clip over stiff material 0.125 inch thick. Repeat this operation 25 times. Then slide the clip over stiff material 0.016 inch thick and invert so that only the tension of the clip is supporting the full weight of the marker. Check for compliance with 3.5.6.

4.3.4.1.7 Startability. Write 4 five-letter words with an unused sample marker. Then place marker on its side with cap off for three hours. Immediately draw the marker nib across a sheet of paper at a rate of approximately 4 inches per second, in increments of 8 inches, until a solid continuous mark at least 0.015 inch wide is attained. Examine markings for compliance with 3.5.7. Then place marker horizontally with cap off for an additional 15 hours and recap for 30 minutes. Remove the cap and draw the marker nib across the paper as described above and examine the markings for compliance with 3.5.7.

4.3.4.1.8 Accelerated aging. Draw an 8-inch linear mark with an unused marker. Recap the marker tightly and place horizontally in a convection-type oven at a constant temperature of 60° ± 1° C. (140° ± 1.8° F.) for 7 days. At the end of the aging period remove the marker from the oven and examine it for leaks and distortion. Then place the marker on its side and allow it to cool at room temperature for 24 hours. After the 24 hour period, remove the cap and draw a mark parallel to the existing mark and check for compliance with 3.5.8.

4.3.4.1.9 Machine write-out test. Select the marker tested in 4.3.4.1.8 and mount in writing position on the machine (see 4.3.4.1). Set the machine speed at 11 feet per minute and the writing pressure at 50 grams. Operate the apparatus, stopping for a 15-minute recuperation period after each successive 300 feet of writing. Continue until the marker fails at any point during writing to produce a solid continuous line 0.015 to 0.050 inch wide. At this point stop the apparatus and check for compliance with 3.5.9. (Note: During each 15-minute recuperation period rotate the marker 90° in the holding device and recap until the next "write" period.)

4.3.4.1.10 Light fastness. Cut a minimum length of four inches of acceptable (meeting test requirements) writing from a portion of write-out prepared in 4.3.4.1.9 after the first 300 feet. Expose the specimen to the radiation from the carbon-arc lamp (see 4.3.4.1). Exposure periods shall be: 24 hours for black, green, orange, and yellow markings; 12 hours for red, blue, purple, and brown markings. Examine for compliance with 3.5.10.

4.3.4.1.11 Body and cap tightness. Select three unused sample markers. Uncap and recap each marker to a normal hand tightness 20 consecutive times. Then uncap the markers and draw a 3-foot linear mark with each and recap tightly. Immediately take and record the total weight of the three markers. Then allow the markers to stand tightly capped for 24 hours. At the end of the 24-hour period, again take and record the total weight of the three markers. The difference of the two total weights divided by 3 shall be considered the average weight loss per marker. Check the caps and the average weight loss for compliance with 3.5.11.

4.3.4.1.12 Impact test. Select one of the markers tested in the previous test, making sure the cap is in place. Drop the marker on a wooden floor 5 times from a height of 4 feet. Position the marker so it will hit the floor at the end equipped with the writing nib. After the fifth drop examine the inside of the cap and the area around the writing nib for evidence of leakage. Examine the cap and barrel for defects. Then write a few sharp turns, reversals, and circles and check for compliance with 3.5.12.

4.3.4.2 Material test.

4.3.4.2.1 Flammability test for plastic. Select a marker previously tested and perform the following test on the cap, if plastic; the barrel, if plastic; or both, if both the cap and barrel are plastic. Remove the contents and metal components from the cap and barrel before performing this test. In a draft-free area, hold the plastic specimen horizontally above a bunsen burner having a 1 to 1-1/4 inch blue flame just touching the end of the specimen until it ignites, or for a period not over 30 seconds. Remove the flame. If the specimen will not support combustion, it is nonflammable. If the specimen burns, measure the burning rate in inches per minute and check for compliance with 3.3.1. Failure of either the cap or barrel to pass this test shall be cause for rejection of the lot.

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5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A, B, or C, as specified (see 6.1).

5.1.1 Level A. Twelve markers of one color shall be packaged in a close-fitting box conforming to PPP-B-566, PPP-B-665, or PPP-B-676.

5.1.2 Level B. The markers shall be packaged as specified in 5.1.1.

5.1.3 Level C. The markers shall be packaged to afford adequate protection against corrosion, deterioration, and damage during shipment from the supply source to the first receiving activity for immediate use. The supplier's commercial practice will be acceptable when it fulfills the above requirements.

5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.1).

5.2.1 Level A. Thirty-six unit packages of markers shall be packed in a close-fitting box conforming to PPP-B-636, class weather-resistant. The box shall be closed, waterproof sealed, and strapped in accordance with the appendix to the box specification.

5.2.2 Level B. Thirty-six unit packages of markers shall be packed in a close-fitting box conforming to PPP-B-636, class-domestic. The box shall be closed in accordance with the appendix to the box specification.

5.2.3 Level C. The markers shall be packed to insure carrier acceptance and safe delivery to destination at the lowest rates in containers complying with the rules and regulations applicable to the mode of transportation.

5.3 Marking.

5.3.1 Civil agencies. In addition to markings required by the contract or order, the interior packages and shipping containers shall be marked in accordance with Fed. Std. No. 123.

5.3.2 Military activities. In addition to the markings required by the contract or order, the interior packages and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Color required (see 1.2.1).
(c) Selection of applicable level of packaging and packing and any special marking required (see 5.1, 5.2, and 5.3).

GSA DC 67-8310

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